第七章 重積分

7.1 二重積分

概念與基本性質

二重積分 $\approx$ (帶符號) 體積: z 軸上方為正, 下方為負.

定義. 給定 $\Omega = [a,\, b] \times [c,\, d]$, $b \geqslant a$, $d \geqslant c$, $f : \Omega \to \mathbb{R}$.

- Ω 分割 $\mathbb{P} : a = x_0 < x_1 < x_2 < \cdots < x_n = b,\ c = y_0 < y_1 < y_2 < \cdots < y_m = d$
- $\Delta x_k = x_k - x_{k-1},\ \ \Delta y_l = y_l - y_{l-1},\ \ k = 1,\, 2,\, \ldots,\, n;\ \ l = 1,\, 2,\, \ldots,\, m$
- $\|\mathbb{P}\| = \max\{\Delta x_k \Delta y_l \mid 1 \leqslant k \leqslant n,\ 1 \leqslant l \leqslant m\}$
- 樣本點 (ξ_k, ζ_l): $x_{k-1} \leqslant \xi_k \leqslant x_k,\ \ y_{l-1} \leqslant \zeta_l \leqslant y_l,\ \ k = 1,\, 2,\, \ldots,\, n;\ \ l = 1,\, 2,\, \ldots,\, m.$
- $u_{k,l} = \sup\{f(x,y) \mid x_{k-1} \leqslant x \leqslant x_k,\ y_{l-1} \leqslant y \leqslant y_l\},\ \ \ell_{k,l} = \inf\{f(x,y) \mid x_{k-1} \leqslant x \leqslant x_k,\ y_{l-1} \leqslant y \leqslant y_l\}$, $k = 1,\, 2,\, \ldots,\, n;\ \ l = 1,\, 2,\, \ldots,\, m.$
- $R(f,\mathbb{P}) = \displaystyle\sum_{l=1}^{m}\sum_{k=1}^{n} f(\xi_k,\zeta_l)\,\Delta x_k \Delta y_l,\ \ U(f,\mathbb{P}) = \sum_{l=1}^{m}\sum_{k=1}^{n} u_{k,l}\,\Delta x_k \Delta y_l,\ \ L(f,\mathbb{P}) = \sum_{l=1}^{m}\sum_{k=1}^{n} \ell_{k,l}\,\Delta x_k \Delta y_l;$
  顯然 $L(f,\mathbb{P}) \leqslant R(f,\mathbb{P}) \leqslant U(f,\mathbb{P})$.
- 求 $\displaystyle\lim_{\|\mathbb{P}\|\to 0} R(f,\mathbb{P})$. 若對不同分割與樣本點選取此極限均存在且相等, 稱 f 在 Ω 可積 (分); $f(x,y)$ 在 Ω 的定積分為 $\displaystyle\int_\Omega f(x,y)\,\mathrm{d}A \equiv \lim_{\|\mathbb{P}\|\to 0} R(f,\mathbb{P})$.

性質.

- 若 f 在 Ω 有界, 且除了 Ω 中有限個平滑曲線外 f 在 Ω 連續, 則 f 在 Ω 可積分.
- 若 $\Omega = \Omega_1 \cap \Omega_2$ 且 $\Omega_1,\ \Omega_2$ 均為矩形, f 在 $\Omega_1,\, \Omega_2$ 均可積, 則 $\displaystyle\int_\Omega f\,\mathrm{d}A = \int_{\Omega_1} f\,\mathrm{d}A + \int_{\Omega_2} f\,\mathrm{d}A.$
- 若 $f_1,\ f_2$ 均在 Ω 可積且 $f_1(x,y) \leqslant f_2(x,y)\ \forall\,(x,y) \in \Omega$, 則 $\displaystyle\int_\Omega f_1\,\mathrm{d}A \leqslant \int_\Omega f_2\,\mathrm{d}A.$
- 若 $\alpha,\, \beta \in \mathbb{R},\ f_1,\ f_2$ 均在 Ω 可積, 則 $\displaystyle\int_\Omega (\alpha f_1 + \beta f_2)\,\mathrm{d}A = \alpha\int_\Omega f_1\,\mathrm{d}A + \beta\int_\Omega f_2\,\mathrm{d}A.$

逐次積分

矩形積分區域

定理 (Fubini)**.** 若 $\Omega = [a,\, b] \times [c,\, d]$, $f : \Omega \to \mathbb{R}$ 為可積, 則

$$\int_\Omega f(x,y)\,\mathrm{d}A = \int_c^d \left\{\int_a^b f(x,y)\,\mathrm{d}x\right\}\mathrm{d}y = \int_a^b \left\{\int_c^d f(x,y)\,\mathrm{d}y\right\}\mathrm{d}x$$

例. 令 $\Omega = \{(x,\, y) \mid -1 \leqslant x \leqslant 1,\ -2 \leqslant y \leqslant 2\}$, 則 $\displaystyle\int_\Omega \sqrt{1-x^2}\,\mathrm{d}A = \frac{\pi}{2} \times 4 = 2\pi.$

例. 求 $\displaystyle\int_0^2\int_1^3 x^2 y\,\mathrm{d}y\,\mathrm{d}x$ 與 $\displaystyle\int_1^3\int_0^2 x^2 y\,\mathrm{d}x\,\mathrm{d}y.$

解.

- $\displaystyle\int_0^2\int_1^3 x^2 y\,\mathrm{d}y\,\mathrm{d}x = \int_0^2 x^2\Big(\frac{y^2}{2}\Big|_1^3\Big)\,\mathrm{d}x = \int_0^2 x^2\Big(\frac{9}{2} - \frac{1}{2}\Big)\,\mathrm{d}x = \frac{4}{3}x^3\Big|_0^2 = \frac{32}{3}.$

- $\displaystyle\int_1^3\int_0^2 x^2y\,\mathrm{d}x\,\mathrm{d}y=\int_1^3 y\Big(\frac{x^3}{3}\Big|_0^2\Big)\,\mathrm{d}y=\int_1^3 y\,\frac{8}{3}\,\mathrm{d}x=\frac{4}{3}y^2\Big|_1^3=\frac{36}{3}-\frac{4}{3}=\frac{32}{3}.$

例. 若 $\Omega=[0,\,1]\times[0,\,3]$, 求 $\displaystyle\int_\Omega e^{2x+y}\,\mathrm{d}A$.

解. $\displaystyle\int_\Omega e^{2x+y}\,\mathrm{d}A=\int_0^3\int_0^1 e^{2x+y}\,\mathrm{d}x\,\mathrm{d}y=\int_0^3 e^y\Big(\int_0^1 e^{2x}\,\mathrm{d}x\Big)\,\mathrm{d}y=\int_0^3 e^y\Big(\frac{e^{2x}}{2}\Big|_0^1\Big)\,\mathrm{d}y=\frac{(e^2-1)(e^3-1)}{2}.$

例. 若 $\Omega=[0,\,\pi]\times[0,\,2\pi]$, 求 $\displaystyle\int_\Omega \sin(x+y)\,\mathrm{d}A$.

解.

- $\displaystyle\int_\Omega \sin(x+y)\,\mathrm{d}A=\int_0^\pi\int_0^{2\pi}\sin(x+y)\,\mathrm{d}y\,\mathrm{d}x=\int_0^\pi\Big(-\cos(x+y)\Big|_0^{2\pi}\Big)\,\mathrm{d}x=-\int_0^\pi(\cos(x+2\pi)-\cos x)\,\mathrm{d}x=0.$
- $\displaystyle\int_\Omega \sin(x+y)\,\mathrm{d}A=\int_0^{2\pi}\int_0^\pi\sin(x+y)\,\mathrm{d}x\,\mathrm{d}y=\int_0^{2\pi}\Big(-\cos(x+y)\Big|_0^\pi\Big)\,\mathrm{d}y=-\int_0^{2\pi}(\cos(\pi+y)-\cos y)\,\mathrm{d}y=-\int_0^{2\pi}(\cos\pi\cos y-\sin\pi\sin y-\cos y)\,\mathrm{d}y=2\int_0^{2\pi}\cos y\,\mathrm{d}y=0.$

例. 若 $\Omega=[1,\,2]\times[0,\,1]$, 求 $\displaystyle\int_\Omega ye^{xy}\,\mathrm{d}A$.

解.

- $\displaystyle\int_\Omega ye^{xy}\,\mathrm{d}A=\int_0^1\int_1^2 ye^{xy}\,\mathrm{d}x\,\mathrm{d}y=\int_0^1 y\Big(\frac{e^{xy}}{y}\Big|_1^2\Big)\,\mathrm{d}y=\int_0^1(e^{2y}-e^y)\,\mathrm{d}y=\frac{e^2}{2}-e+\frac{1}{2}.$
- $\displaystyle\int_\Omega ye^{xy}\,\mathrm{d}A=\int_1^2\int_0^1 ye^{xy}\,\mathrm{d}y\,\mathrm{d}x=\int_1^2\Big(y\,\frac{e^{xy}}{x}-\frac{e^{xy}}{x^2}\Big|_0^1\Big)\,\mathrm{d}x=\int_1^2\Big(\frac{e^x}{x}-\frac{e^x}{x^2}+\frac{1}{x^2}\Big)\,\mathrm{d}x=\Big(\frac{e^x}{x}-\frac{1}{x}\Big)\Big|_1^2=\frac{e^2}{2}-e+\frac{1}{2}.$

習題. 求下列重積分.

1. $\displaystyle\int_2^3\int_1^5(x+2y)\,\mathrm{d}x\,\mathrm{d}y$
2. $\displaystyle\int_0^{\frac{\pi}{2}}\int_0^{\cos y}e^x\sin y\,\mathrm{d}x\,\mathrm{d}y$
3. $\displaystyle\int_{-\pi}^{\pi}\int_0^2 r\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta$
4. $\displaystyle\int_0^{\ln 4}\int_0^{\ln 3}e^{x+y}\,\mathrm{d}x\,\mathrm{d}y$
5. $\displaystyle\int_0^{\frac{\pi}{2}}\int_0^{\cos\theta}\rho^2\sin\theta\,\mathrm{d}\rho\,\mathrm{d}\theta$
6. $\displaystyle\int_0^1\int_{x^2}^{x^3}(x^2+y^2)\,\mathrm{d}y\,\mathrm{d}x$
7. $\displaystyle\int_1^2\int_0^y x\sqrt{y^2-x^2}\,\mathrm{d}x\,\mathrm{d}y$
8. $\displaystyle\int_0^1\int_{y^4}^{y^2}\sqrt{\frac{y}{x}}\,\mathrm{d}x\,\mathrm{d}y$

解.

1. $\displaystyle\int_2^3\int_1^5(x+2y)\,\mathrm{d}x\,\mathrm{d}y=\int_2^3\Big(\frac{x^2}{2}+2yx\Big)\Big|_1^5\,\mathrm{d}y=\int_2^3(12+8y)\,\mathrm{d}y=(12y+4y^2)\Big|_2^3=32$
2. $\displaystyle\int_{-\pi}^{\pi}\int_0^2 r\sin\theta\,\mathrm{d}r\,\mathrm{d}\theta=\Big(\int_{-\pi}^{\pi}\sin\theta\,\mathrm{d}\theta\Big)\cdot\Big(\int_0^2 r\,\mathrm{d}r\Big)=0$
3. $\displaystyle\int_0^{\ln 4}\int_0^{\ln 3}e^{x+y}\,\mathrm{d}x\,\mathrm{d}y=\Big(\int_0^{\ln 4}e^y\,\mathrm{d}y\Big)\cdot\Big(\int_0^{\ln 3}e^x\,\mathrm{d}x\Big)=(4-1)(3-1)=6$
4. $\displaystyle\int_0^{\frac{\pi}{2}}\int_0^{\cos y}e^x\sin y\,\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{2}}\sin y\Big(\int_0^{\cos y}e^x\,\mathrm{d}x\Big)\,\mathrm{d}y=\int_0^{\frac{\pi}{2}}\sin y\,(e^{\cos y}-1)\,\mathrm{d}y=\big(-e^{\cos y}+\cos y\big)\Big|_0^{\frac{\pi}{2}}=e-2$

5. $\displaystyle\int_0^{\frac{\pi}{2}}\int_0^{\cos\theta}\rho^2\sin\theta\,\mathrm{d}\rho\,\mathrm{d}\theta=\int_0^{\frac{\pi}{2}}\sin\theta\left(\int_0^{\cos\theta}\rho^2\,\mathrm{d}\rho\right)\mathrm{d}\theta=\int_0^{\frac{\pi}{2}}\sin\theta\,\frac{\cos^3\theta}{3}\,\mathrm{d}\theta=-\frac{\cos^4\theta}{12}\bigg|_0^{\frac{\pi}{2}}=\frac{1}{12}$

6. $\displaystyle\int_0^1\int_{x^2}^{x^3}(x^2+y^2)\,\mathrm{d}y\,\mathrm{d}x=\int_0^1\left(x^2y+\frac{y^3}{3}\right)\bigg|_{y=x^2}^{y=x^3}\mathrm{d}x=\int_0^1\left(x^2\cdot x^3+\frac{x^9}{3}-x^2\cdot x^2-\frac{x^6}{3}\right)\mathrm{d}x$
$\displaystyle=\left(\frac{x^6}{6}+\frac{x^{10}}{30}-\frac{x^5}{5}-\frac{x^7}{21}\right)\bigg|_0^1=-\frac{1}{21}$

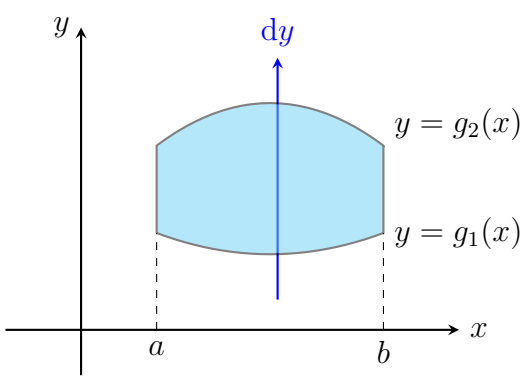
7. $\displaystyle\int_0^y x\sqrt{y^2-x^2}\,\mathrm{d}x$ 中, 令 $u=y^2-x^2$, 則 $\mathrm{d}u=-2x\,\mathrm{d}x\implies x\,\mathrm{d}x=\frac{-1}{2}\,\mathrm{d}u$. 積分範圍 x 由 0 至 y, 則變數變換 $u=y^2-x^2$ 由 $y^2-0^2=y^2$ 至 $y^2-y^2=0$. 則 $\displaystyle\int_0^y x\sqrt{y^2-x^2}\,\mathrm{d}x=\int_{y^2}^0\sqrt{u}\,\frac{-1}{2}\,\mathrm{d}u=\frac{1}{2}\int_0^{y^2}\sqrt{u}\,\mathrm{d}u=\frac{1}{2}\,\frac{2}{3}u^{\frac{3}{2}}\bigg|_0^{y^2}=\frac{y^3}{3}$. 故 $\displaystyle\int_1^2\int_0^y x\sqrt{y^2-x^2}\,\mathrm{d}x\,\mathrm{d}y=\int_1^2\frac{y^3}{3}\,\mathrm{d}y=\frac{y^4}{12}\bigg|_1^2=\frac{5}{4}$.

8. $\displaystyle\int_0^1\int_{y^4}^{y^2}\sqrt{\frac{y}{x}}\,\mathrm{d}x\,\mathrm{d}y=\int_0^1\sqrt{y}\left(\int_{y^4}^{y^2}\frac{1}{\sqrt{x}}\,\mathrm{d}x\right)\mathrm{d}y=\int_0^1\sqrt{y}\,(2\sqrt{x})\bigg|_{y^4}^{y^2}\mathrm{d}y=2\int_0^1\left(y^{\frac{3}{2}}-y^{\frac{5}{2}}\right)\mathrm{d}y$
$\displaystyle=2\left(\frac{2}{5}y^{\frac{5}{2}}-\frac{2}{7}y^{\frac{7}{2}}\right)\bigg|_0^1=\frac{8}{35}$

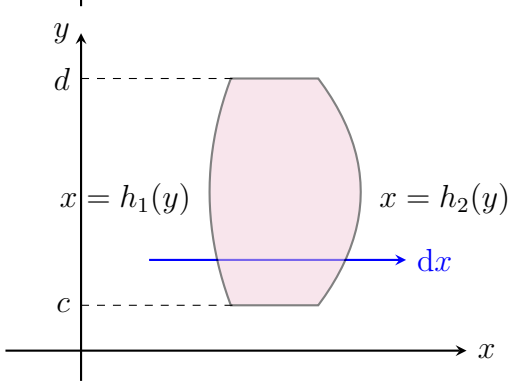
一般積分區域

結論 (基本區域積分).

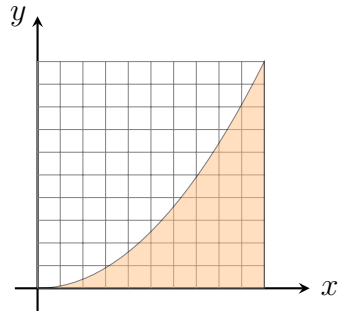
1. 選擇積分方向 v: 延 x 方向為 $\mathrm{d}x$, 延 y 方向為 $\mathrm{d}y$.
2. 想像積分方向 v 為一射線, 進入區域之邊界函數為積分下界, 離開區域之邊界函數為積分上界; 邊界函數本身不含變數 v.
3. 次第積分區域為原始積分區域延 v 之投影, 亦即令 $v=0$ 之方程式; 積分方向選擇如上步驟.



$$\int_\Omega f(x,y)\,\mathrm{d}A=\int_a^b\int_{g_1(x)}^{g_2(x)}f(x,y)\,\mathrm{d}y\,\mathrm{d}x$$

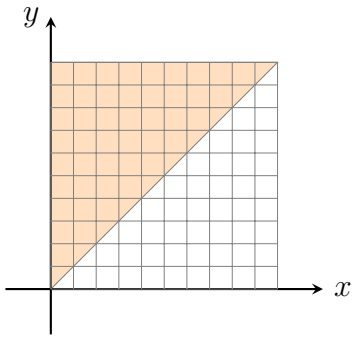


$$\int_\Omega f(x,y)\,\mathrm{d}A=\int_c^d\int_{h_1(y)}^{h_2(y)}f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$



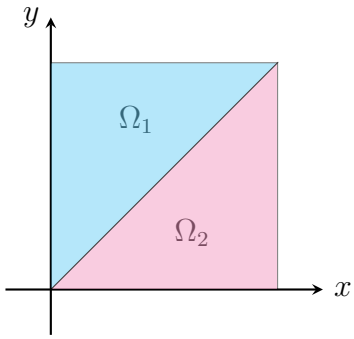
例. 求 $\displaystyle\int_\Omega x\cos y\,\mathrm{d}A$, Ω 為 $y=0$, $x=1$, 與 $y=x^2$ 圍成之區域.

解. $\displaystyle\int_\Omega x\cos y\,\mathrm{d}A=\int_0^1\int_0^{x^2}x\cos y\,\mathrm{d}y\,\mathrm{d}x=\int_0^1 x\left(\sin y\Big|_0^{x^2}\right)\mathrm{d}x$
$\displaystyle=\int_0^1 x\sin x^2\,\mathrm{d}x=\frac{1-\cos 1}{2}$



例. 求 $\displaystyle\int_\Omega e^{-y^2}\,\mathrm{d}A$, $\Omega=\{(x,\,y)\,|\,0\leqslant y\leqslant 1,\ 0\leqslant x\leqslant y\}$.

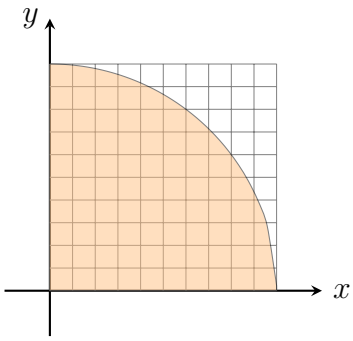
解. $\displaystyle\int_\Omega e^{-y^2}\,\mathrm{d}A=\int_0^1\int_0^y e^{-y^2}\,\mathrm{d}x\,\mathrm{d}y=\int_0^1 ye^{-y^2}\,\mathrm{d}y=\frac{1-e^{-1}}{2}$



例. 求 $\displaystyle\int_0^1\int_0^1 e^{\max\{x^2,\,y^2\}}\,\mathrm{d}x\,\mathrm{d}y$.

解. $\displaystyle\int_0^1\int_0^1 e^{\max\{x^2,\,y^2\}}\,\mathrm{d}x\,\mathrm{d}y=\int_{\Omega_1}e^{y^2}\,\mathrm{d}A+\int_{\Omega_2}e^{x^2}\,\mathrm{d}A$
$\displaystyle=\int_0^1\int_0^y e^{y^2}\,\mathrm{d}x\,\mathrm{d}y+\int_0^1\int_0^x e^{x^2}\,\mathrm{d}y\,\mathrm{d}x=\int_0^1 ye^{y^2}\,\mathrm{d}y+\int_0^1 xe^{x^2}\,\mathrm{d}x=e-1.$

積分順序交換



例. 求 $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\sqrt{1-y^2}\,\mathrm{d}y\,\mathrm{d}x$.

解. $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\sqrt{1-y^2}\,\mathrm{d}y\,\mathrm{d}x=\int_0^1\int_0^{\sqrt{1-y^2}}\sqrt{1-y^2}\,\mathrm{d}x\,\mathrm{d}y=\int_0^1(1-y^2)\,\mathrm{d}y=\frac{2}{3}.$
$\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\sqrt{1-y^2}\,\mathrm{d}y\,\mathrm{d}x=\frac{1}{2}\int_0^1(\sin^{-1}\sqrt{1-x^2}+x\sqrt{1-x^2})\,\mathrm{d}x=\frac{1}{2}\left(1+\frac{1}{3}\right)=\frac{2}{3}.$

例. 將下列積分以不同積分順序寫出.

1. $\displaystyle\int_0^2\int_{x^2}^{2x}f(x,y)\,\mathrm{d}y\,\mathrm{d}x$
2. $\displaystyle\int_0^1\int_{\sin^{-1}y}^{\frac{\pi}{2}}f(x,y)\,\mathrm{d}x\,\mathrm{d}y$
3. $\displaystyle\int_{-1}^2\int_{x^2}^{x+2}f(x,y)\,\mathrm{d}y\,\mathrm{d}x$
4. $\displaystyle\int_0^1\int_0^{2y}f(x,y)\,\mathrm{d}x\,\mathrm{d}y+\int_1^3\int_0^{3-y}f(x,y)\,\mathrm{d}x\,\mathrm{d}y$

解.

1. $\displaystyle\int_0^2\int_{x^2}^{2x}f(x,y)\,\mathrm{d}y\,\mathrm{d}x=\int_0^4\int_{\sqrt{y}}^{\frac{y}{2}}f(x,y)\,\mathrm{d}x\,\mathrm{d}y$
2. $\displaystyle\int_0^1\int_{\sin^{-1}y}^{\frac{\pi}{2}}f(x,y)\,\mathrm{d}x\,\mathrm{d}y=\int_0^{\frac{\pi}{2}}\int_0^{\sin x}f(x,y)\,\mathrm{d}x\,\mathrm{d}y$

3. $\displaystyle\int_{-1}^{2}\int_{x^2}^{x+2} f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_0^1\int_{-\sqrt{y}}^{\sqrt{y}} f(x,y)\,\mathrm{d}x\,\mathrm{d}y + \int_1^4\int_{y-2}^{\sqrt{y}} f(x,y)\,\mathrm{d}x\,\mathrm{d}y$

4. $\displaystyle\int_0^1\int_0^{2y} f(x,y)\,\mathrm{d}x\,\mathrm{d}y + \int_1^3\int_0^{3-y} f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_0^2\int_{\frac{x}{2}}^{3-x} f(x,y)\,\mathrm{d}y\,\mathrm{d}x$

例. 證明 $\displaystyle\int_0^x\int_0^t F(u)\,\mathrm{d}u\,\mathrm{d}t = \int_0^x (x-u)F(u)\,\mathrm{d}u.$

解. $\displaystyle\int_0^x\int_0^t F(u)\,\mathrm{d}u\,\mathrm{d}t = \int_0^x\int_u^x F(u)\,\mathrm{d}t\,\mathrm{d}u = \int_0^x (x-u)F(u)\,\mathrm{d}u.$

例. 求 $\displaystyle\int_0^1\int_x^1 e^{-y^2}\,\mathrm{d}y\,\mathrm{d}x.$

解. $\displaystyle\int_0^1\int_x^1 e^{-y^2}\,\mathrm{d}y\,\mathrm{d}x = \int_0^1\int_0^y e^{-y^2}\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 ye^{-y^2}\,\mathrm{d}y = \frac{1-e^{-1}}{2}$

例. 求 $\displaystyle\int_0^1\int_y^1 \frac{\sin x}{x}\,\mathrm{d}x\,\mathrm{d}y.$

解. $\displaystyle\int_0^1\int_y^1 \frac{\sin x}{x}\,\mathrm{d}x\,\mathrm{d}y = \int_0^1\int_0^x \frac{\sin x}{x}\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 \sin x\,\mathrm{d}x = 1-\cos 1.$

例. 求 $\displaystyle\int_0^1\int_{\sin^{-1}y}^{\frac{\pi}{2}} \cos x\sqrt{1+\cos^2 x}\,\mathrm{d}x\,\mathrm{d}y.$

解.

- $\displaystyle\int_0^1\int_{\sin^{-1}y}^{\frac{\pi}{2}} \cos x\sqrt{1+\cos^2 x}\,\mathrm{d}x\,\mathrm{d}y = \int_0^{\frac{\pi}{2}}\int_0^{\sin x} \cos x\sqrt{1+\cos^2 x}\,\mathrm{d}y\,\mathrm{d}x = \int_0^{\frac{\pi}{2}} \sin x\cos x\sqrt{1+\cos^2 x}\,\mathrm{d}x$
  $\displaystyle= \int_0^1 u\sqrt{1+u^2}\,\mathrm{d}u = \int_1^2 \sqrt{v}\,\frac{1}{2}\,\mathrm{d}v = \frac{2\sqrt{2}-1}{3}$, 其中變數變換 $u=\cos x$ 與 $v=1+u^2$.
- 令 $u=\sin x$, $\displaystyle\int_{\sin^{-1}y}^{\frac{\pi}{2}} \cos x\sqrt{1+\cos^2 x}\,\mathrm{d}x = \int_y^1 \sqrt{2-u^2}\,\mathrm{d}u$. 令 $u=\sqrt{2}\sin\theta$, 則 $\displaystyle\int_y^1 \sqrt{2-u^2}\,\mathrm{d}u =$
  $\displaystyle\int_{\sin^{-1}\frac{y}{\sqrt{2}}}^{\frac{\pi}{4}} 2\cos^2\theta\,\mathrm{d}\theta = \int_{\sin^{-1}\frac{y}{\sqrt{2}}}^{\frac{\pi}{4}} (1+\cos 2\theta)\,\mathrm{d}\theta = (\theta+\sin\theta\cos\theta)\Big|_{\sin^{-1}\frac{y}{\sqrt{2}}}^{\frac{\pi}{4}} = \frac{\pi}{4}+\frac{1}{2}-\sin^{-1}\frac{y}{\sqrt{2}}-\frac{y}{\sqrt{2}}\cdot\frac{\sqrt{2-y^2}}{\sqrt{2}} =$
  $\displaystyle\frac{\pi}{4}+\frac{1}{2}-\sin^{-1}\frac{y}{\sqrt{2}}-\frac{y\sqrt{2-y^2}}{2}$. 故 $\displaystyle\int_0^1\int_{\sin^{-1}y}^{\frac{\pi}{2}} \cos x\sqrt{1+\cos^2 x}\,\mathrm{d}x\,\mathrm{d}y$
  $\displaystyle= \int_0^1\left(\frac{\pi}{4}+\frac{1}{2}-\sin^{-1}\frac{y}{\sqrt{2}}-\frac{y\sqrt{2-y^2}}{2}\right)\mathrm{d}y = \frac{\pi}{4}+\frac{1}{2}-\left(\frac{\pi}{4}+1-\sqrt{2}\right)-\left(\frac{\sqrt{2}}{3}-\frac{1}{6}\right) = \frac{2\sqrt{2}-1}{3}.$

例. 求 $\displaystyle\int_0^\infty \frac{e^{-ax}-e^{-bx}}{x}\,\mathrm{d}x,\ b>a>0.$

解. 由 $\displaystyle\int_a^b e^{-xy}\,\mathrm{d}y = \frac{1}{-x}e^{-xy}\Big|_{y=a}^{y=b} = \frac{1}{-x}(e^{-xb}-e^{-xa}) = \frac{e^{-ax}-e^{-bx}}{x}$, $\displaystyle\int_0^\infty \frac{e^{-ax}-e^{-bx}}{x}\,\mathrm{d}x = \int_0^\infty\left(\int_a^b e^{-xy}\,\mathrm{d}y\right)\mathrm{d}x =$
$\displaystyle\int_a^b\left(\int_0^\infty e^{-xy}\,\mathrm{d}x\right)\mathrm{d}y = \int_a^b \frac{1}{y}\,\mathrm{d}y = \ln\frac{b}{a}.$

例. 求 $\displaystyle\int_0^\infty \frac{\tan^{-1}\pi x-\tan^{-1}x}{x}\,\mathrm{d}x.$

解. 由 $\int_x^{\pi x} \frac{1}{1+y^2}\,\mathrm{d}y = \tan^{-1}\pi x - \tan^{-1} x$, $\int_0^\infty \frac{\tan^{-1}\pi x - \tan^{-1} x}{x}\,\mathrm{d}x = \int_0^\infty \frac{1}{x}\Big(\int_x^{\pi x} \frac{1}{1+y^2}\,\mathrm{d}y\Big)\,\mathrm{d}x$
$= \int_0^\infty \int_x^{\pi x} \frac{1}{x\,(1+y^2)}\,\mathrm{d}y\,\mathrm{d}x = \int_0^\infty \int_{\frac{y}{\pi}}^{y} \frac{1}{x\,(1+y^2)}\,\mathrm{d}x\,\mathrm{d}y = \int_0^\infty \frac{1}{1+y^2}\Big(\int_{\frac{y}{\pi}}^{y} \frac{1}{x}\,\mathrm{d}x\Big)\,\mathrm{d}y = \int_0^\infty \frac{1}{1+y^2}\Big(\ln y - \ln\frac{y}{\pi}\Big)\,\mathrm{d}y =$
$\int_0^\infty \frac{\ln\pi}{1+y^2}\,\mathrm{d}y = \frac{\pi}{2}\ln\pi.$

例. 求 $\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x.$

解. 由 $\int_0^\infty e^{-xy}\,\mathrm{d}y = \frac{1}{x}$, $x>0$; 又 $\int e^{ax}\sin bx\,\mathrm{d}x = \frac{e^{ax}(a\cdot\sin bx - b\cdot\cos bx)}{a^2+b^2}$.
故 $\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \int_0^\infty \sin x\Big(\int_0^\infty e^{-xy}\,\mathrm{d}y\Big)\,\mathrm{d}x = \int_0^\infty\Big(\int_0^\infty e^{-xy}\sin x\,\mathrm{d}y\Big)\,\mathrm{d}x = \int_0^\infty\Big(\int_0^\infty e^{-xy}\sin x\,\mathrm{d}x\Big)\,\mathrm{d}y =$
$\int_0^\infty \frac{e^{-xy}((-y)\cdot\sin x - 1\cdot\cos x)}{(-y)^2+1^2}\Big|_{x=0}^{x=\infty}\,\mathrm{d}y = \int_0^\infty \frac{0-(-1)}{1+y^2}\,\mathrm{d}y = \int_0^\infty \frac{1}{1+y^2}\,\mathrm{d}y = \tan^{-1} y\Big|_0^\infty = \frac{\pi}{2}.$

例 (Fubini 定理之反例)**.** 求 $\int_0^1\int_0^1 \frac{x-y}{(x+y)^3}\,\mathrm{d}y\,\mathrm{d}x$ 與 $\int_0^1\int_0^1 \frac{x-y}{(x+y)^3}\,\mathrm{d}x\,\mathrm{d}y.$

解.

- 令 $t=x+y$, 則 $y=t-x$, $\mathrm{d}y=\mathrm{d}t$, y 由 0 到 1, $t=x+y$ 由 $x+0=x$ 到 $x+1$, $\int_0^1 \frac{x-y}{(x+y)^3}\,\mathrm{d}y = \int_x^{x+1} \frac{2x-t}{t^3}\,\mathrm{d}t = \Big(-\frac{x}{t^2}+\frac{1}{t}\Big)\Big|_x^{x+1} = \frac{1}{(x+1)^2}$; 故 $\int_0^1\int_0^1 \frac{x-y}{(x+y)^3}\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 \frac{1}{(x+1)^2}\,\mathrm{d}x = \frac{1}{2}.$
- 令 $t=x+y$, 則 $x=t-y$, $\mathrm{d}x=\mathrm{d}t$, x 由 0 到 1, $t=x+y$ 由 $y+0=y$ 到 $y+1$, $\int_0^1 \frac{x-y}{(x+y)^3}\,\mathrm{d}x = \int_y^{y+1} \frac{t-2y}{t^3}\,\mathrm{d}t = \Big(-\frac{1}{t}+\frac{y}{t^2}\Big)\Big|_y^{y+1} = -\frac{1}{(y+1)^2}$; 故 $\int_0^1\int_0^1 \frac{x-y}{(x+y)^3}\,\mathrm{d}x\,\mathrm{d}y = -\int_0^1 \frac{1}{(y+1)^2}\,\mathrm{d}x = -\frac{1}{2}.$

變數變換

單變數積分 $\int_{[a,\,b]} f(x)\,\mathrm{d}x$ 中, 令 $x=x(u)$, 則 $\mathrm{d}x = \frac{\mathrm{d}x}{\mathrm{d}u}\cdot\mathrm{d}u$; 故 $\int_{[a,\,b]} f(x)\,\mathrm{d}x = \int_{x^{-1}[a,b]} f(x(u))\cdot\frac{\mathrm{d}x}{\mathrm{d}u}\cdot\mathrm{d}u.$

例.

- 求 $\int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^2}}$: 令 $x=\sin u$, 則 $\mathrm{d}x=\cos u\cdot\mathrm{d}u$; x 由 0 至 1, 則 u 由 $\sin^{-1}0=0$ 至 $\sin^{-1}1=\frac{\pi}{2}$. 故 $\int_0^1 \frac{\mathrm{d}x}{\sqrt{1-x^2}} = \int_0^{\frac{\pi}{2}} \frac{\cos u\cdot\mathrm{d}u}{\sqrt{1-\sin^2 u}} = \int_0^{\frac{\pi}{2}} \frac{\cos u\cdot\mathrm{d}u}{\cos u} = \int_0^{\frac{\pi}{2}} 1\cdot\mathrm{d}u = \frac{\pi}{2}.$
- 求 $\int_{\frac{1}{2}}^1 (2x-1)^2\,\mathrm{d}x$: 令 $u=2x-1$, 亦即 $x=\frac{u}{2}+\frac{1}{2}$, 則 $\mathrm{d}x=\frac{1}{2}\,\mathrm{d}u$; x 由 $\frac{1}{2}$ 至 1, 則 u 由 $2\cdot\frac{1}{2}-1=0$ 至 $2\cdot 1-1=1$. 故 $\int_{\frac{1}{2}}^1 (2x-1)^2\,\mathrm{d}x = \int_0^1 u^2\,\frac{1}{2}\,\mathrm{d}u = \frac{1}{6}.$

定理 (重積分變數變換)**.** 給定 Ω_x, $\Omega_u \subseteq \mathbb{R}^n$, $\mathbf{x}=\mathbf{x}(\mathbf{u}) : \Omega_u \to \Omega_x$ 且

- $\mathbf{x}$ 為嵌射 (亦即 $\mathbf{x}(\Omega_u)=\Omega_x$, $\mathbf{x}^{-1}(\Omega_x)=\Omega_u$)
- $\mathbf{x}$ 之各分量函數 x_j, $j=1,\,2\,\ldots,\,n$ 為連續可偏微分

- $\forall\, \mathbf{u} \in \Omega_u$, $\mathbf{x}$ 之 Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_n} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_n} & \cdots & \frac{\partial x_2}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \frac{\partial x_n}{\partial u_n} & \cdots & \frac{\partial x_n}{\partial u_n} \end{vmatrix} (\mathbf{u}) \neq 0$

則對於可積函數 $f : \Omega_x \to \mathbb{R}$, $\displaystyle\int_{\Omega_x} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_{\mathbf{x}^{-1}(\Omega_x)} f(\mathbf{x}(\mathbf{u}))\,\big|J_{\mathbf{x}}(\mathbf{u})\big|\,\mathrm{d}\mathbf{u} = \int_{\mathbf{x}^{-1}(\Omega_x)} f(\mathbf{x}(\mathbf{u}))\,\left|\dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}\right|\mathrm{d}\mathbf{u}$.

例. 利用變數變換 $x = u^2 - v^2$, $y = 2uv$ 求 $\displaystyle\int_\Omega y\,\mathrm{d}A$, Ω 為 $y \geqslant 0$, $y^2 = 4 - 4x$, $y^2 = 4 + 4x$ 所圍成之區域.

解. 由 $\begin{cases} x = u^2 - v^2 \\ y = 2uv \end{cases}$, Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4\,(u^2 + v^2)$. 變數變換 $(x,\, y) \to (u,\, v)$ 後 Ω 由 $uv = 0$, $u^2v^2 + u^2 - v^2 - 1 = (v^2+1)(u^2-1) = 0$, $u^2v^2 - u^2 + v^2 - 1 = (u^2+1)(v^2-1) = 0$ 圍成, 亦即 $\Omega = \{0 \leqslant u \leqslant 1,\ 0 \leqslant v \leqslant 1\}$, 故 $\displaystyle\int_\Omega y\,\mathrm{d}A = \int_0^1\!\!\int_0^1 2uv\,\Big|4\,(u^2+v^2)\Big|\,\mathrm{d}u\,\mathrm{d}v = 2$.

例. 求 $\displaystyle\int_\Omega (x+y)^2\,\mathrm{d}A$, Ω 為 $x + y = 0$, $x + y = 1$, $2x - y = 0$, $2x - y = 3$ 所圍成之平行四邊形.

解. 令 $\begin{cases} u = x + y \\ v = 2x - y \end{cases}$, 則 $\begin{cases} x = \frac{1}{3}u + \frac{1}{3}v \\ y = \frac{2}{3}u - \frac{1}{3}v \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{-1}{3} \end{vmatrix} = -\dfrac{1}{3}$. 變數變換 $(x,\, y) \to (u,\, v)$ 後 $\Omega = \{0 \leqslant u \leqslant 1,\ 0 \leqslant v \leqslant 3\}$, 故 $\displaystyle\int_\Omega (x+y)^2\,\mathrm{d}A = \int_0^3\!\!\int_0^1 u^2\,\Big|-\frac{1}{3}\Big|\,\mathrm{d}u\,\mathrm{d}v = \frac{1}{3}$.

例. 求 $\displaystyle\int_\Omega \sqrt{x+y}\,\mathrm{d}A$, Ω 為頂點 $(0,\, 0)$, $(a,\, 0)$, $(0,\, a)$, $a > 0$ 之三角形.

解. 令 $\begin{cases} u = x + y \\ v = x - y \end{cases}$, 則 $\begin{cases} x = \frac{1}{2}u + \frac{1}{2}v \\ y = \frac{1}{2}u - \frac{1}{2}v \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{-1}{2} \end{vmatrix} = -\dfrac{1}{2}$. 變數變換前 Ω 由 $x + y = 0$, $x + y = a$, $x = 0$, $y = 0$ 所圍成, 變數變換 $(x,\, y) \to (u,\, v)$ 後 Ω 由 $u = 0$, $u = a$, $u + v = 0$, $u - v = 0$ 圍成, 故 $\displaystyle\int_\Omega \sqrt{x+y}\,\mathrm{d}A = \int_0^a\!\!\int_{-u}^u \sqrt{u}\,\Big|-\frac{1}{2}\Big|\,\mathrm{d}v\,\mathrm{d}u = \int_0^a u\sqrt{u}\,\mathrm{d}u = \dfrac{2\,a^{\frac{5}{2}}}{5}$.

例. 求 $\displaystyle\int_\Omega (x+y)\,e^{x-y}\,\mathrm{d}A$, Ω 為頂點 $(4,\, 0)$, $(6,\, 2)$, $(4,\, 4)$, $(2,\, 2)$ 之四邊形.

解. 令 $\begin{cases} u = x + y \\ v = x - y \end{cases}$, 則 $\begin{cases} x = \frac{1}{2}u + \frac{1}{2}v \\ y = \frac{1}{2}u - \frac{1}{2}v \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{-1}{2} \end{vmatrix} = -\dfrac{1}{2}$. 變數變換前 Ω 由 $x - y = 0$, $x - y = 4$, $x + y = 4$, $x + y = 8$ 所圍成, 變數變換 $(x,\, y) \to (u,\, v)$ 後 Ω 由 $v = 0$, $v = 4$, $u = 4$, $u = 8$ 圍成, 故 $\displaystyle\int_\Omega (x+y)\,e^{x-y}\,\mathrm{d}A = \int_0^4\!\!\int_4^8 ue^v\,\Big|-\frac{1}{2}\Big|\,\mathrm{d}u\,\mathrm{d}v = \frac{1}{2}\int_0^4 e^v\,\mathrm{d}v\int_4^8 u\,\mathrm{d}u = 12\,(e^4 - 1)$.

例. 求 $\displaystyle\int_\Omega e^{\frac{x+y}{x-y}}\,\mathrm{d}A$, Ω 為頂點 $(1,\, 0)$, $(2,\, 0)$, $(0,\, -2)$, $(0,\, -1)$ 之梯形.

解. 令 $\begin{cases} u = x + y \\ v = x - y \end{cases}$, 則 $\begin{cases} x = \frac{1}{2}u + \frac{1}{2}v \\ y = \frac{1}{2}u - \frac{1}{2}v \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{-1}{2} \end{vmatrix} = -\dfrac{1}{2}$. 變數變換前 Ω 由 $x - y = 1$, $x - y = 2$, $x = 0$, $y = 0$ 所圍成, 變數變換 $(x,\, y) \to (u,\, v)$ 後 Ω 由 $v = 1$, $v = 2$, $u + v = 0$, $u - v = 0$ 圍成, 故 $\displaystyle\int_\Omega e^{\frac{x+y}{x-y}}\,\mathrm{d}A = \int_1^2\!\!\int_{-v}^v e^{\frac{u}{v}}\,\Big|-\frac{1}{2}\Big|\,\mathrm{d}u\,\mathrm{d}v = \frac{1}{2}\int_1^2 \Big(v\,e^{\frac{u}{v}}\Big|_{u=-v}^{u=v}\Big)\,\mathrm{d}v = \dfrac{3\,(e - e^{-1})}{4}$.

例. 求 $y=x^2$, $y=2x^2$, $x=y^2$, $x=3y^2$ 所圍成之區域面積.

解. 令 $\begin{cases} u=\frac{y}{x^2} \\ v=\frac{x}{y^2} \end{cases}$, 則 $\begin{cases} x=(u^2v)^{-\frac{1}{3}} \\ y=(uv^2)^{-\frac{1}{3}} \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u})=\dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u})=\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}=\begin{vmatrix} -\frac{1}{3}(u^2v)^{-\frac{4}{3}}\cdot 2uv & -\frac{1}{3}(u^2v)^{-\frac{4}{3}}\cdot u^2 \\ -\frac{1}{3}(uv^2)^{-\frac{4}{3}}\cdot v^2 & -\frac{1}{3}(uv^2)^{-\frac{4}{3}}\cdot 2uv \end{vmatrix}$ $=\dfrac{1}{3u^2v^2}$. 變數變換 $(x,\,y)\to(u,\,v)$ 後 Ω 由 $u=1$, $u=2$, $v=1$, $v=3$ 圍成, 亦即 $\Omega=\{1\leqslant u\leqslant 2,\ 1\leqslant v\leqslant 3\}$, 面積為 $\displaystyle\int_\Omega \mathrm{d}A=\int_1^3\int_1^2\left|\frac{1}{3u^2v^2}\right|\mathrm{d}u\,\mathrm{d}v=\frac{1}{9}$.

例. 求 $\displaystyle\int_\Omega \frac{xy}{1+x^2y^2}\,\mathrm{d}A$, Ω 為 $xy=1$, $xy=5$, $x=1$, $x=6$ 所圍成之區域.

解. 令 $\begin{cases} u=xy \\ v=x \end{cases}$, 則 $\begin{cases} x=v \\ y=\frac{u}{v} \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u})=\dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u})=\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}=\begin{vmatrix} 0 & 1 \\ \frac{1}{v} & -\frac{u}{v^2} \end{vmatrix}=-\dfrac{1}{v}$. 變數變換 $(x,\,y)\to(u,\,v)$ 後 Ω 由 $u=1$, $u=5$, $v=1$, $v=6$ 圍成, 亦即 $\Omega=\{1\leqslant u\leqslant 5,\ 1\leqslant v\leqslant 6\}$, 故 $\displaystyle\int_\Omega \frac{xy}{1+x^2y^2}\,\mathrm{d}A=\int_1^6\int_1^5 \frac{u}{1+u^2}\left|-\frac{1}{v}\right|\mathrm{d}u\,\mathrm{d}v=\int_1^6\frac{1}{v}\,\mathrm{d}v\int_1^5\frac{u}{1+u^2}\,\mathrm{d}u=\frac{\ln 6\cdot\ln 13}{2}$.

例. 求 $\displaystyle\int_\Omega \frac{y}{x}\,\mathrm{d}A$, Ω 為 $y=x$, $x^2+4y^2=4$, $y\geqslant 0$ 所圍成之區域.

解. 令 $\begin{cases} x=2r\cos\theta \\ y=r\sin\theta \end{cases}$, Jacobian $J_{\mathbf{x}}(\mathbf{u})=\dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u})=\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}=\begin{vmatrix} 2\cos\theta & -2r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}=2r$. 變數變換 $(x,\,y)\to(r,\,\theta)$ 後 Ω 由 $r=0$, $r=1$, $\theta=0$, $\theta=\tan^{-1}2$ 圍成, 亦即 $\Omega=\{0\leqslant r\leqslant 1,\ 0\leqslant\theta\leqslant\tan^{-1}2\}$, 故 $\displaystyle\int_\Omega\frac{y}{x}\,\mathrm{d}A=\int_0^{\tan^{-1}2}\int_0^1\frac{1}{2}\,\tan\theta\,\big|2r\big|\,\mathrm{d}r\,\mathrm{d}\theta=\int_0^{\tan^{-1}2}\tan\theta\,\mathrm{d}\theta\int_0^1 r\,\mathrm{d}r=-\frac{1}{2}\,\ln|\cos\theta|\Big|_0^{\tan^{-1}2}=\frac{\ln 5}{4}$.

例. 求 $\displaystyle\int_\Omega \sin(9x^2+4y^2)\,\mathrm{d}A$, Ω 為 $9x^2+4y^2=1$, $y\geqslant 0$, $x\geqslant 0$ 所圍成之區域.

解. 令 $\begin{cases} x=\frac{1}{3}\,r\cos\theta \\ y=\frac{1}{2}\,r\sin\theta \end{cases}$, Jacobian $J_{\mathbf{x}}(\mathbf{u})=\dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u})=\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}=\begin{vmatrix} \frac{1}{3}\,\cos\theta & -\frac{1}{3}\,r\sin\theta \\ \frac{1}{2}\sin\theta & \frac{1}{2}\,r\cos\theta \end{vmatrix}=\dfrac{1}{6}\,r$. 變數變換 $(x,\,y)\to(r,\,\theta)$ 後 Ω 由 $r=0$, $r=1$, $\theta=0$, $\theta=\dfrac{\pi}{2}$ 圍成, 亦即 $\Omega=\{0\leqslant r\leqslant 1,\ 0\leqslant\theta\leqslant\dfrac{\pi}{2}\}$, 故 $\displaystyle\int_\Omega\sin(9x^2+4y^2)\,\mathrm{d}A=\int_0^{\frac{\pi}{2}}\int_0^1\sin r^2\,\Big|\frac{1}{6}\,r\Big|\,\mathrm{d}r\,\mathrm{d}\theta=\frac{1}{6}\int_0^{\frac{\pi}{2}}\mathrm{d}\theta\int_0^1 r\sin r^2\,\mathrm{d}r=\frac{\pi}{24}(1-\cos 1)$.

例. 求 $\displaystyle\int_0^1\int_0^{1-x}\sqrt{x+y}\,(y-2x)^2\,\mathrm{d}y\,\mathrm{d}x$.

解. 令 $\begin{cases} u=x+y \\ v=y-2x \end{cases}$, 則 $\begin{cases} x=\frac{1}{3}u-\frac{1}{3}v \\ y=\frac{2}{3}u+\frac{1}{3}v \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u})=\dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u})=\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}=\begin{vmatrix} \frac{1}{3} & \frac{-1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix}=\dfrac{1}{3}$. 變數變換前積分區域由 $x+y=0$, $x+y=1$, $x=0$, $y=0$ 所圍成, 變數變換 $(x,\,y)\to(u,\,v)$ 後積分區域由 $u=0$, $u=1$, $u-v=0$, $2u+v=0$ 圍成, 故 $\displaystyle\int_0^1\int_0^{1-x}\sqrt{x+y}\,(y-2x)^2\,\mathrm{d}y\,\mathrm{d}x=\int_0^1\int_{-2u}^{u}\sqrt{u}\,v^2\left|\frac{1}{3}\right|\mathrm{d}v\,\mathrm{d}u=\frac{1}{3}\int_0^1\sqrt{u}\left(\frac{v^3}{3}\Big|_{v=-2u}^{v=u}\right)\mathrm{d}u=\frac{2}{9}$.

例. 若 Ω 為頂點 $(0,\,0)$, $(1,\,0)$, $(0,\,1)$ 之三角形, f 為可積, 證明 $\displaystyle\int_\Omega f(x+y)\,\mathrm{d}A=\int_0^1 u\,f(u)\,\mathrm{d}u$.

解. 令 $\begin{cases} u = x+y \\ v = x \end{cases}$, 則 $\begin{cases} x = v \\ y = u-v \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$. 變數變換前 Ω 由 $x+y=0$, $x+y=1$, $x=0$, $y=0$ 所圍成, 變數變換 $(x,\,y) \to (u,\,v)$ 後 Ω 由 $u=0$, $u=1$, $v=0$, $u-v=0$ 圍成, 故 $\displaystyle\int_\Omega f(x+y)\,\mathrm{d}A = \int_0^1\int_0^u f(u)\,\big|-1\big|\,\mathrm{d}v\,\mathrm{d}u = \int_0^1 u\,f(u)\,\mathrm{d}u$.

例. 1. 證明 $\displaystyle\int_0^1\int_0^1 \frac{1}{1-xy}\,\mathrm{d}x\,\mathrm{d}y = \sum_{n=1}^\infty \frac{1}{n^2}$. 2. 以變數變換證明 $\displaystyle\int_0^1\int_0^1 \frac{1}{1-xy}\,\mathrm{d}x\,\mathrm{d}y = \frac{\pi^2}{6}$.

解.

1. $|xy| < 1$ 時 $\displaystyle\frac{1}{1-xy} = \sum_{n=0}^\infty (xy)^n$ 且可逐項積分; 代入 $\displaystyle\int_0^1\int_0^1 \frac{1}{1-xy}\,\mathrm{d}x\,\mathrm{d}y = \int_0^1\int_0^1 \sum_{n=0}^\infty (xy)^n\,\mathrm{d}x\,\mathrm{d}y = \sum_{n=0}^\infty \int_0^1 x^n\,\mathrm{d}x \int_0^1 y^n\,\mathrm{d}y = \sum_{n=0}^\infty \frac{1}{(n+1)^2} = \sum_{n=1}^\infty \frac{1}{n^2}$.
2. 令 $\begin{cases} x = \frac{u-v}{\sqrt{2}} \\ y = \frac{u+v}{\sqrt{2}} \end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{vmatrix} = 1$. 變數變換前積分區域由 $x=0$, $x=1$, $y=0$, $y=1$ 所圍成, 變數變換 $(x,\,y) \to (u,\,v)$ 後積分區域由 $u-v=0$, $u-v=\sqrt{2}$, $u+v=0$, $u+v=\sqrt{2}$ 圍成, 故 $\displaystyle\int_0^1\int_0^1 \frac{1}{1-xy}\,\mathrm{d}x\,\mathrm{d}y = \int_0^{\frac{\sqrt{2}}{2}}\int_{-u}^{u} \frac{\mathrm{d}v\,\mathrm{d}u}{1-\frac{u-v}{\sqrt{2}}\frac{u+v}{\sqrt{2}}} + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}}\int_{-\sqrt{2}+u}^{\sqrt{2}-u} \frac{\mathrm{d}v\,\mathrm{d}u}{1-\frac{u-v}{\sqrt{2}}\frac{u+v}{\sqrt{2}}} = 2\left(\int_0^{\frac{\sqrt{2}}{2}}\int_{-u}^{u} \frac{\mathrm{d}v\,\mathrm{d}u}{2-u^2+v^2} + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}}\int_{-\sqrt{2}+u}^{\sqrt{2}-u} \frac{\mathrm{d}v\,\mathrm{d}u}{2-u^2+v^2}\right) = 2\left(\int_0^{\frac{\sqrt{2}}{2}} \frac{1}{\sqrt{2-u^2}}\left(\tan^{-1}\frac{v}{\sqrt{2-u^2}}\Bigg|_{v=-u}^{v=u}\right)\mathrm{d}u + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}} \frac{1}{\sqrt{2-u^2}}\left(\tan^{-1}\frac{v}{\sqrt{2-u^2}}\Bigg|_{v=-\sqrt{2}+u}^{v=\sqrt{2}-u}\right)\mathrm{d}u\right) = 4\left(\int_0^{\frac{\sqrt{2}}{2}} \frac{1}{\sqrt{2-u^2}}\tan^{-1}\frac{u}{\sqrt{2-u^2}}\,\mathrm{d}u + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}} \frac{1}{\sqrt{2-u^2}}\tan^{-1}\frac{\sqrt{2}-u}{\sqrt{2-u^2}}\,\mathrm{d}u\right)$. 令 $u = \sqrt{2}\sin\theta$, 則 $\mathrm{d}u = \sqrt{2}\cos\theta\,\mathrm{d}\theta$; 化簡 $4\left(\displaystyle\int_0^{\frac{\sqrt{2}}{2}} \frac{1}{\sqrt{2-u^2}}\tan^{-1}\frac{u}{\sqrt{2-u^2}}\,\mathrm{d}u + \int_{\frac{\sqrt{2}}{2}}^{\sqrt{2}} \frac{1}{\sqrt{2-u^2}}\tan^{-1}\frac{\sqrt{2}-u}{\sqrt{2-u^2}}\,\mathrm{d}u\right)$ $= 4\left(\displaystyle\int_0^{\frac{\pi}{6}} \tan^{-1}(\tan\theta)\,\mathrm{d}\theta + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \tan^{-1}\left(\frac{1-\sin\theta}{\cos\theta}\right)\mathrm{d}\theta\right) = 4\left(\int_0^{\frac{\pi}{6}} \theta\,\mathrm{d}\theta + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1}{2}\left(\frac{\pi}{2}-\theta\right)\mathrm{d}\theta\right) = \frac{\pi^2}{6}$; 其中 $\dfrac{1-\sin\theta}{\cos\theta} = \dfrac{1-\cos\left(\frac{\pi}{2}-\theta\right)}{\sin\left(\frac{\pi}{2}-\theta\right)} = \dfrac{2\,\sin^2\frac{1}{2}\left(\frac{\pi}{2}-\theta\right)}{2\,\sin\frac{1}{2}\left(\frac{\pi}{2}-\theta\right)\cos\frac{1}{2}\left(\frac{\pi}{2}-\theta\right)} = \tan\dfrac{1}{2}\left(\dfrac{\pi}{2}-\theta\right)$.

極座標二重積分

令 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$, Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \dfrac{\partial \mathbf{x}}{\partial \mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$.

例. 求 $\displaystyle\int_\Omega (3x+4y^2)\,\mathrm{d}A$, Ω 為 $x^2+y^2=1$, $x^2+y^2=4$ 與 $y \geqslant 0$ 所圍成之區域.

解. 變數變換 $(x,\,y) \to (r,\,\theta)$ 後 Ω 由 $r=1$, $r=2$, $\theta=0$, $\theta=\pi$ 圍成, 故 $\displaystyle\int_\Omega (3x+4y^2)\,\mathrm{d}A = \int_0^\pi\int_1^2 (3r\cos\theta + 4r^2\sin^2\theta)\,r\,\mathrm{d}r\,\mathrm{d}\theta = \int_0^\pi (7\,\cos\theta + 15\,\sin^2\theta)\,\mathrm{d}\theta = \frac{15\,\pi}{2}$.

例. 求 $z = 1-x^2-y^2$ 與 $z \geqslant 0$ 所圍成之體積.

解. 體積為 $\int_\Omega (1-x^2-y^2)\,\mathrm{d}A$, Ω 為單位圓 $x^2+y^2=1$. 變數變換 $(x,\,y)\to(r,\,\theta)$ 後 Ω 由 $r=0$, $r=1$, $\theta=0$, $\theta=2\pi$ 圍成, 故 $\int_\Omega (1-x^2-y^2)\,\mathrm{d}A=\int_0^{2\pi}\int_0^1 (1-r^2)\,r\,\mathrm{d}r\,\mathrm{d}\theta=\dfrac{\pi}{2}$.

例. 求 $\int_\Omega \dfrac{y^2}{x^2}\,\mathrm{d}A$, Ω 為 $0<a^2\leqslant x^2+y^2\leqslant b^2$ 與 $x\geqslant y\geqslant 0$ 所圍成之區域.

解. 變數變換 $(x,\,y)\to(r,\,\theta)$ 後 Ω 由 $r=a$, $r=b$, $\theta=0$, $\theta=\dfrac{\pi}{4}$ 圍成, 故 $\int_\Omega \dfrac{y^2}{x^2}\,\mathrm{d}A=\int_0^{\frac{\pi}{4}}\int_a^b \tan^2\theta\cdot r\,\mathrm{d}r\,\mathrm{d}\theta=\dfrac{b^2-a^2}{2}\int_0^{\frac{\pi}{4}}\tan^2\theta\,\mathrm{d}\theta=\dfrac{b^2-a^2}{2}\int_0^{\frac{\pi}{4}}(\sec^2\theta-1)\,\mathrm{d}\theta=\dfrac{b^2-a^2}{2}(\tan\theta-\theta)\Big|_0^{\frac{\pi}{4}}=\dfrac{b^2-a^2}{2}\Big(1-\dfrac{\pi}{4}\Big)$.

例. 求 $\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x$.

解. $\Big(\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x\Big)^2=\Big(\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x\Big)\Big(\int_{-\infty}^{\infty} e^{-y^2}\,\mathrm{d}y\Big)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2-y^2}\,\mathrm{d}x\,\mathrm{d}y$. 變數變換 $(x,\,y)\to(r,\,\theta)$ 後得 $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2-y^2}\,\mathrm{d}x\,\mathrm{d}y=\int_0^{2\pi}\int_0^{\infty} e^{-r^2}r\,\mathrm{d}r\,\mathrm{d}\theta=2\pi\left(\dfrac{-e^{-r^2}}{2}\Big|_0^{\infty}\right)=\pi$, 故 $\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x=\sqrt{\pi}$.

例 (Γ 函數).

1. 證明 $\Gamma(x)\equiv\int_0^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂, $\forall\, x>0$.
2. 證明 $\Gamma(x+1)=x\,\Gamma(x)$, $\forall\, x>0$.
3. 證明 $\Gamma(n+1)=n!$, $\forall\, n\in\mathbb{N}$.
4. 求 $\Gamma\Big(\dfrac{1}{2}\Big)$ 與 $\Gamma\Big(\dfrac{3}{2}\Big)$.
5. 求 $\int_0^{\infty} t\,e^{-\alpha\,t^2}\,\mathrm{d}t$ 與 $\int_0^{\infty} t^2\,e^{-\alpha\,t^2}\,\mathrm{d}t$, $\alpha>0$.

解.

1. (證一) 由 $\lim\limits_{t\to\infty} t^{x-1}e^{-\frac{t}{2}}=0$, $\exists\, T>0$ 使 $t^{x-1}e^{-\frac{t}{2}}\leqslant 1\ \forall\, t\geqslant T$. $\int_0^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t=\int_0^{T} t^{x-1}e^{-t}\,\mathrm{d}t+\int_T^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t\leqslant\int_0^{T} t^{x-1}\,\mathrm{d}t+\int_T^{\infty} 1\cdot e^{-\frac{t}{2}}\,\mathrm{d}t=\dfrac{T^x}{x}+2\,e^{-\frac{T}{2}}$. 故 $\forall\, x>0$, $\int_0^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂.

   (證二) $\int_0^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t=\int_0^{1} t^{x-1}e^{-t}\,\mathrm{d}t+\int_1^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t$. $\int_1^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂, 因為

   - $\int_0^{1} t^{x-1}e^{-t}\,\mathrm{d}t\leqslant\int_0^{1} t^{x-1}\,\mathrm{d}t=\dfrac{t^x}{x}\Big|_{x=0}^{x=1}=\dfrac{1}{x}$, $\int_0^{1} t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂.
   - $\int_1^{\infty}\dfrac{1}{t^2}\,\mathrm{d}t$ 收斂, $\lim\limits_{t\to\infty}\dfrac{t^{x-1}e^{-t}}{\frac{1}{t^2}}=\lim\limits_{t\to\infty}\dfrac{t^{x+1}}{e^t}=0$, $\int_1^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t$ 收斂.

2. $\Gamma(x+1)=\int_0^{\infty} t^x e^{-t}\,\mathrm{d}t=\lim\limits_{\substack{a\to 0+\\ b\to\infty}}\int_a^b t^x e^{-t}\,\mathrm{d}t$. 令 $u=t^x$, 則 $\mathrm{d}u=xt^{x-1}\,\mathrm{d}t$. 令 $\mathrm{d}v=e^{-t}\,\mathrm{d}t$, 則 $v=-e^{-t}$. 故 $\lim\limits_{\substack{a\to 0+\\ b\to\infty}}\int_a^b t^x e^{-t}\,\mathrm{d}t=\lim\limits_{\substack{a\to 0+\\ b\to\infty}}\Big(-t^x e^{-t}\Big|_a^b+\int_a^b e^{-t}\cdot xt^{x-1}\,\mathrm{d}t\Big)=0+x\int_0^{\infty} t^{x-1}e^{-t}\,\mathrm{d}t=x\,\Gamma(x)$.

3. 由上 $\Gamma(n+1)=n\,\Gamma(n)=n(n-1)\,\Gamma(n-2)=\cdots=n(n-1)\cdots 2\,\Gamma(1)$, 又 $\Gamma(1)=\lim\limits_{b\to\infty}\int_0^b e^{-t}\,\mathrm{d}t=\lim\limits_{b\to\infty}-e^{-t}\Big|_0^b=\lim\limits_{b\to\infty}1-e^{-b}=1$, 得證.

4. (a) $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t}\,\mathrm{d}t = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t$. 令 $u = \sqrt{t}$, 則 $t = u^2$, $\mathrm{d}u = \frac{1}{2\sqrt{t}}\,\mathrm{d}t \implies \frac{1}{\sqrt{t}}\,\mathrm{d}t = 2\,\mathrm{d}u$. 積分範圍 t 由 0 至 ∞, 則變數變換後 u 由 $\sqrt{0} = 0$ 至 $\sqrt{\infty} = \infty$, 故 $\Gamma\left(\frac{1}{2}\right) = \int_0^\infty \frac{e^{-t}}{\sqrt{t}}\,\mathrm{d}t = 2\int_0^\infty e^{-u^2}\,\mathrm{d}u = 2\cdot\frac{\sqrt{\pi}}{2} = \sqrt{\pi}$.

   (b) 由 $\Gamma(x+1) = x\,\Gamma(x)$, $\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2}+1\right) = \frac{1}{2}\,\Gamma\left(\frac{1}{2}\right) = \frac{1}{2}\sqrt{\pi}$.

5. (a) 令 $u = \alpha\,t^2$, 則 $\mathrm{d}u = 2\,\alpha\,t\,\mathrm{d}t \implies t\,\mathrm{d}t = \frac{\mathrm{d}u}{2\alpha}$. 積分範圍 t 由 0 至 ∞, 則變數變換後 u 由 $\alpha\cdot 0^2 = 0$ 至 $\alpha\cdot\infty^2 = \infty$, 故 $\int_0^\infty t\,e^{-\alpha\,t^2}\,\mathrm{d}t = \int_0^\infty e^{-u}\,\frac{\mathrm{d}u}{2\,\alpha} = \frac{1}{2\,\alpha}\int_0^\infty e^{-u}\,\mathrm{d}u = \frac{1}{2\,\alpha}\left(-e^{-u}\Big|_0^\infty\right) = \frac{1}{2\,\alpha}$.

   (b) 令 $u = \alpha\,t^2$, 則 $t = \frac{\sqrt{u}}{\sqrt{\alpha}}$, $\mathrm{d}u = 2\,\alpha\,t\,\mathrm{d}t \implies t\,\mathrm{d}t = \frac{\mathrm{d}u}{2\,\alpha}$. 積分範圍 t 由 0 至 ∞, 則變數變換後 u 由 $\alpha\cdot 0^2 = 0$ 至 $\alpha\cdot\infty^2 = \infty$, 故 $\int_0^\infty t^2\,e^{-\alpha\,t^2}\,\mathrm{d}t = \int_0^\infty t\,e^{-\alpha\,t^2}\,t\,\mathrm{d}t = \int_0^\infty \frac{\sqrt{u}}{\sqrt{\alpha}}\,e^{-u}\,\frac{\mathrm{d}u}{2\,\alpha} = \frac{1}{2\,\alpha^{\frac{3}{2}}}\int_0^\infty u^{\frac{1}{2}}\,e^{-u}\,\mathrm{d}u = \frac{1}{2\,\alpha^{\frac{3}{2}}}\cdot\Gamma\left(\frac{3}{2}\right) = \frac{1}{2\,\alpha^{\frac{3}{2}}}\cdot\frac{1}{2}\sqrt{\pi} = \frac{\sqrt{\pi}}{4\,\alpha^{\frac{3}{2}}}$.

例. 已知 $\forall\, 0 < \alpha < 1$, $\int_0^\infty \frac{x^{\alpha-1}}{1+x}\,\mathrm{d}x = \frac{\pi}{\sin\alpha\pi}$, 證明 $\Gamma(\alpha)\,\Gamma(1-\alpha) = \frac{\pi}{\sin\alpha\pi}$.

解. $\Gamma(\alpha)\,\Gamma(1-\alpha) = \int_0^\infty e^{-t}t^{\alpha-1}\,\mathrm{d}t\int_0^\infty e^{-s}s^{-\alpha}\,\mathrm{d}s = \int_0^\infty\int_0^\infty e^{-(t+s)}t^{\alpha-1}s^{-\alpha}\,\mathrm{d}s\,\mathrm{d}t$. 令 $\begin{cases} u = s+t \\ v = \frac{t}{s}\end{cases}$, 則 $\begin{cases} s = \frac{u}{1+v} \\ t = \frac{uv}{1+v}\end{cases}$; Jacobian $J_{\mathbf{x}}(\mathbf{u}) = \frac{\partial\mathbf{x}}{\partial\mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial s}{\partial u} & \frac{\partial s}{\partial v} \\ \frac{\partial t}{\partial u} & \frac{\partial t}{\partial v}\end{vmatrix} = \begin{vmatrix} \frac{1}{1+v} & \frac{-u}{(1+v)^2} \\ \frac{v}{1+v} & \frac{u}{(1+v)^2}\end{vmatrix} = \frac{u}{(1+v)^2}$. 變數變換 $(s,\,t) \to (u,\,v)$ 後積分範圍仍為 $0 < u < \infty$, $0 < v < \infty$, 故 $\int_0^\infty\int_0^\infty e^{-(t+s)}t^{\alpha-1}s^{-\alpha}\,\mathrm{d}s\,\mathrm{d}t = \int_0^\infty\int_0^\infty e^{-(t+s)}\left(\frac{t}{s}\right)^\alpha t^{-1}\,\mathrm{d}s\,\mathrm{d}t = \int_0^\infty\int_0^\infty e^{-u}v^\alpha\frac{1+v}{uv}\frac{u}{(1+v)^2}\,\mathrm{d}u\,\mathrm{d}v = \int_0^\infty\int_0^\infty e^{-u}\frac{v^{\alpha-1}}{1+v}\,\mathrm{d}u\,\mathrm{d}v = \int_0^\infty e^{-u}\,\mathrm{d}u\int_0^\infty\frac{v^{\alpha-1}}{1+v}\,\mathrm{d}v = \int_0^\infty\frac{v^{\alpha-1}}{1+v}\,\mathrm{d}v = \frac{\pi}{\sin\alpha\pi}$.

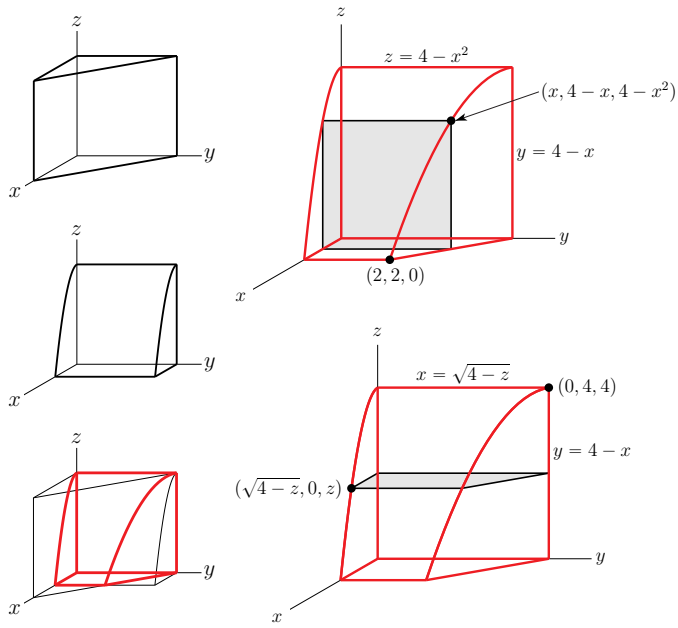
7.2 三重積分

一般區域積分

積分順序交換

例. 若 E 為 $x = 0$, $y = 0$, $z = 0$, $y = 4 - x$, $z = 4 - x^2$ 圍成之區域, 寫出 $\int_E f(x,y,z)\,\mathrm{d}V$ 依積分順序 $z \to y \to x$ 與 $y \to x \to z$ 之積分式.

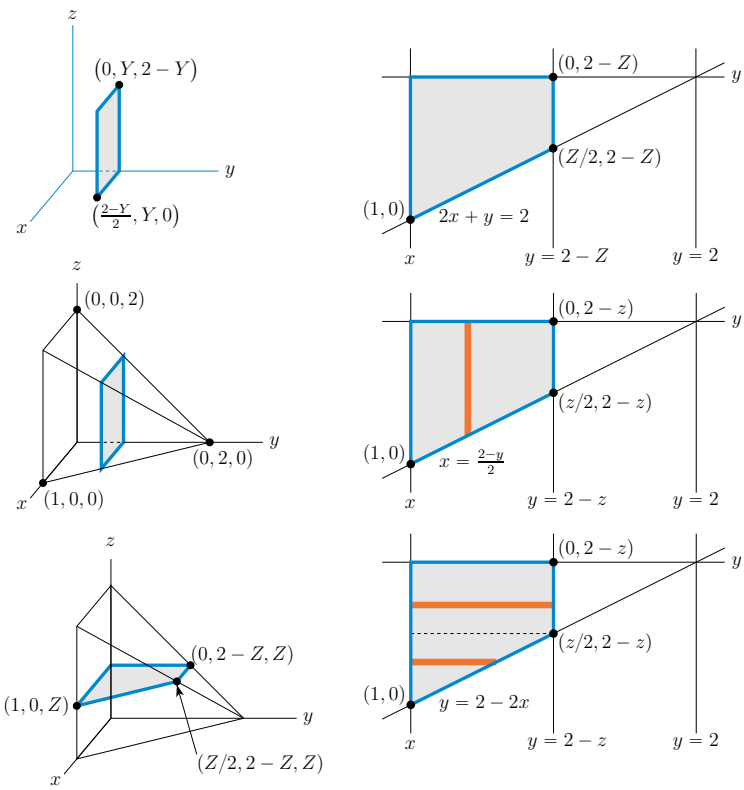
解.



$$\int_E f(x,y,z)\,\mathrm{d}V = \int_0^2\int_0^{4-x}\int_0^{4-x^2} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$$
$$= \int_0^4\int_0^{\sqrt{4-z}}\int_0^{4-x} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}x\,\mathrm{d}z$$

例. 寫出 $\displaystyle\int_0^2\int_0^{2-y}\int_0^{\frac{2-y}{2}} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}z\,\mathrm{d}y$ 依積分順序 $x \to y \to z$ 與 $y \to x \to z$ 之積分式.

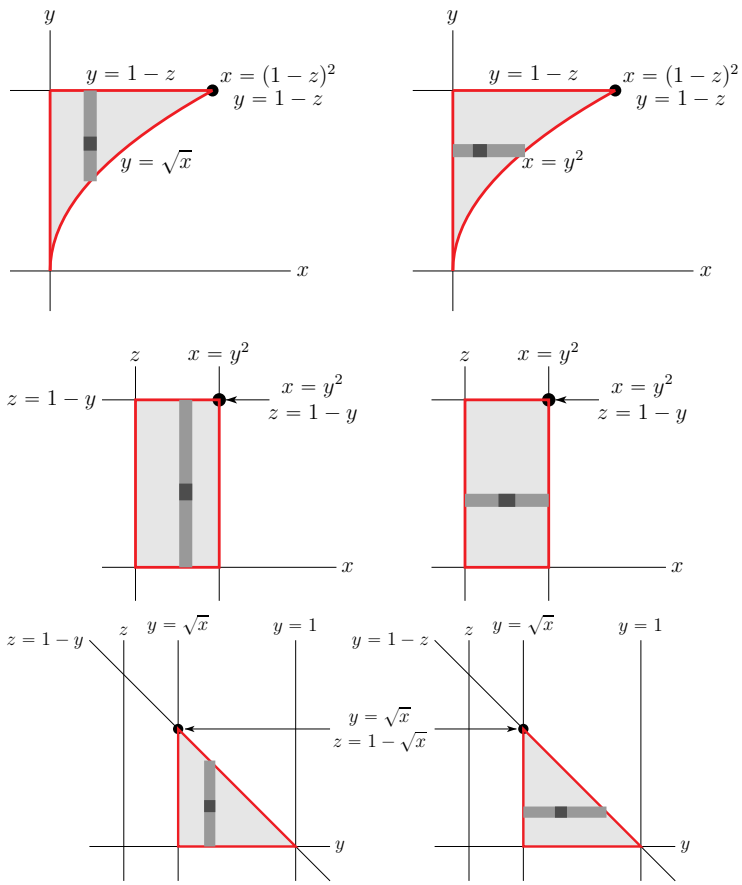
解.



$$\int_0^2\int_0^{2-y}\int_0^{\frac{2-y}{2}} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}z\,\mathrm{d}y$$
$$= \int_0^2\int_0^{2-z}\int_0^{\frac{2-y}{2}} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$$
$$= \int_0^2\int_0^{\frac{z}{2}}\int_0^{2-z} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}x\,\mathrm{d}z$$
$$+ \int_0^2\int_{\frac{z}{2}}^1\int_0^{2-2x} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}x\,\mathrm{d}z$$

例. 若 $\displaystyle\int_E f(x,y,z)\,\mathrm{d}V = \int_0^1\int_{\sqrt{x}}^1\int_0^{1-y} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$, 寫出其他五個等價但積分順序不同之積分式.

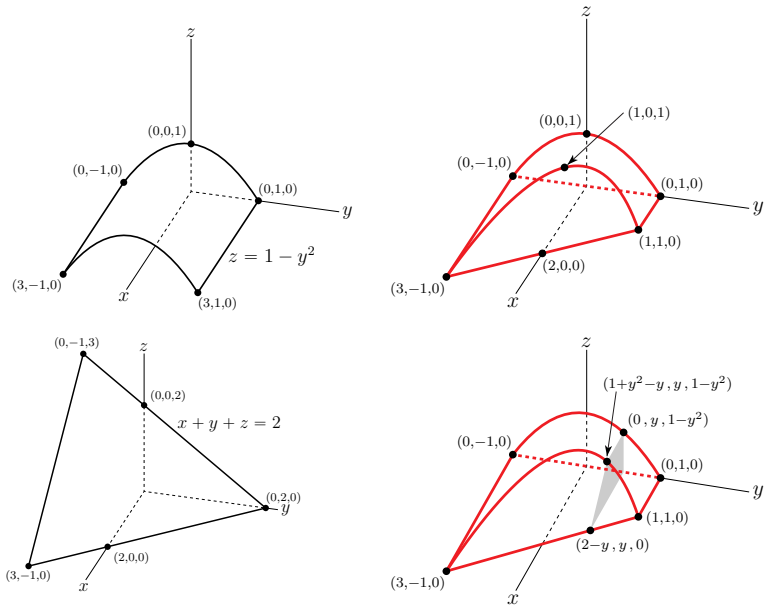
解.



$$
\begin{aligned}
&\int_E f(x,y,z)\,\mathrm{d}V \\
&= \int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 \int_0^{y^2} \int_0^{1-y} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_0^1 \int_0^{1-\sqrt{x}} \int_{\sqrt{x}}^{1-z} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}z\,\mathrm{d}x \\
&= \int_0^1 \int_0^{(1-z)^2} \int_{\sqrt{x}}^{1-z} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}x\,\mathrm{d}z \\
&= \int_0^1 \int_0^{1-y} \int_0^{y^2} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}z\,\mathrm{d}y \\
&= \int_0^1 \int_0^{1-z} \int_0^{y^2} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z
\end{aligned}
$$

例. 若 $\displaystyle\int_E f(x,y,z)\,\mathrm{d}V = \int_{-1}^1 \int_0^{1-y^2} \int_0^{2-y-z} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}z\,\mathrm{d}y$, 寫出等價依積分順序 $z \to x \to y$ 之積分式.

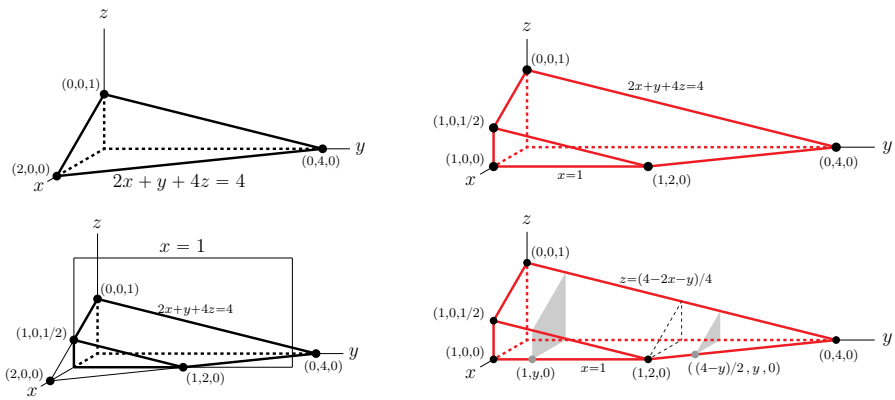
解.



$$
\begin{aligned}
&\int_E f(x,y,z)\,\mathrm{d}V \\
&= \int_{-1}^1 \int_0^{1+y^2-y} \int_0^{1-y^2} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y \\
&\quad + \int_{-1}^1 \int_{1+y^2-y}^{2-y} \int_0^{2-x-y} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y
\end{aligned}
$$

例. 若 $\displaystyle\int_E f(x,y,z)\,\mathrm{d}V = \int_0^1 \int_0^{1-\frac{x}{2}} \int_0^{4-2x-4z} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}z\,\mathrm{d}x$, 寫出等價依積分順序 $z \to x \to y$ 之積分式.

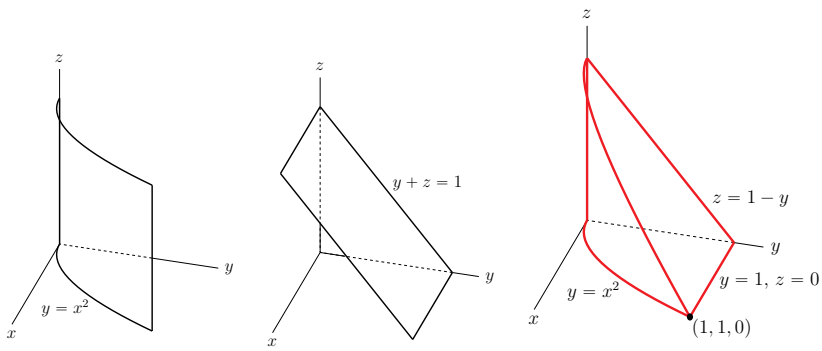
解.



$$\int_E f(x,y,z)\,\mathrm{d}V = \int_0^2\int_0^1\int_0^{\frac{4-2x-y}{2}} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y + \int_2^4\int_0^{\frac{4-y}{2}}\int_0^{\frac{4-2x-y}{2}} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y$$

例. 若 E 為第一卦限中由 $y=x^2$, $y+z=1$ 圍成之區域, 求 $\displaystyle\int_E x\,\mathrm{d}V$.

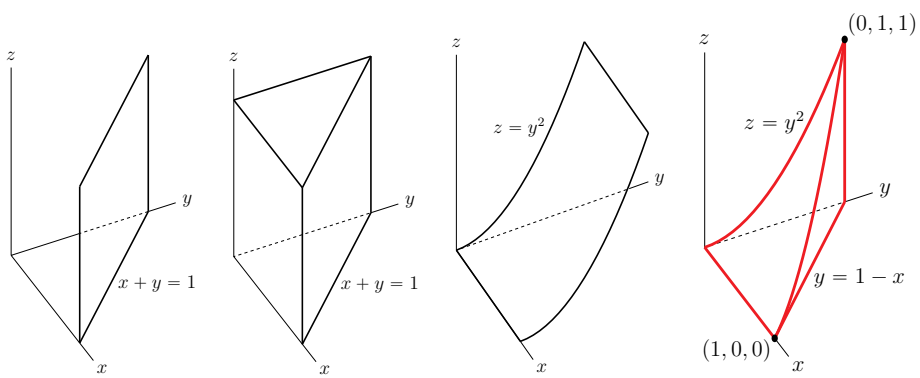
解.



$$\begin{aligned}\int_E x\,\mathrm{d}V &= \int_0^1\int_{x^2}^1\int_0^{1-y} x\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x\\ &= \int_0^1\int_{x^2}^1 x\,(1-y)\,\mathrm{d}y\,\mathrm{d}x\\ &= \int_0^1\left(x(1-x^2) - x\left(\frac{y^2}{2}\Big|_{y=x^2}^{y=1}\right)\right)\mathrm{d}x\\ &= \int_0^1\left(x - x^3 + \frac{x^5}{2} - \frac{x}{2}\right)\mathrm{d}x = \frac{1}{12}\end{aligned}$$

例. 若 E 為第一卦限中由 $z=y^2$, $x+y=1$ 圍成之區域, 求 $\displaystyle\int_E z\,\mathrm{d}V$.

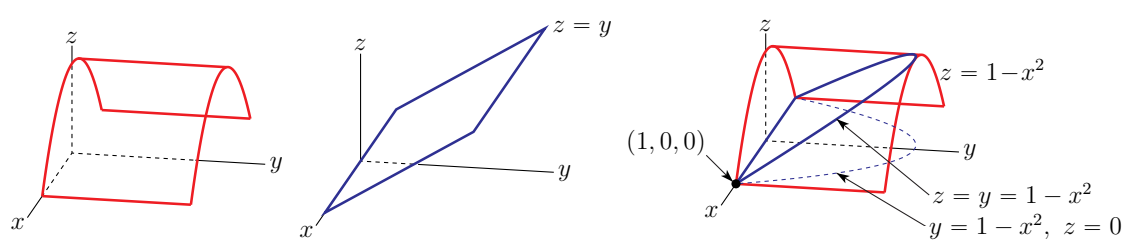
解.



$$\begin{aligned}\int_E z\,\mathrm{d}V &= \int_0^1\int_0^{1-x}\int_0^{y^2} z\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x\\ &= \int_0^1\int_0^{1-x}\frac{y^4}{2}\,\mathrm{d}y\,\mathrm{d}x\\ &= \int_0^1\frac{(1-x)^5}{10}\,\mathrm{d}x = \frac{1}{60}\end{aligned}$$

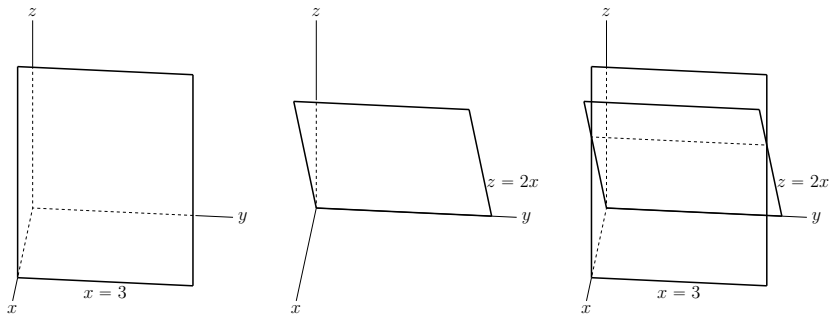
例. 若 E 為 $z=1-x^2$, $y=z$, $y=0$, $z=0$ 圍成之區域, 寫出 $\displaystyle\int_E \mathrm{d}V$ 之積分式.

解.

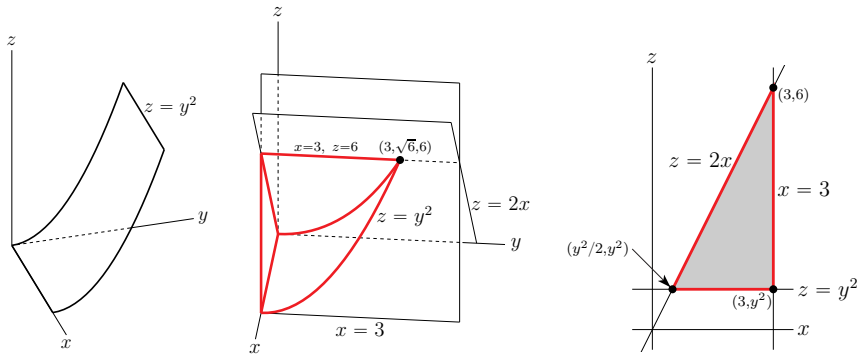


$$\int_E \mathrm{d}V = \int_{-1}^1\int_0^{1-x^2}\int_y^{1-x^2}\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$$

例. 若 E 為 $z=2x$, $z=y^2$, $x=3$ 圍成之區域, 寫出 $\displaystyle\int_E f(x,y,z)\,\mathrm{d}V$ 依積分順序 $z\to x\to y$, $x\to z\to y$, 與 $y\to x\to z$ 之積分式.

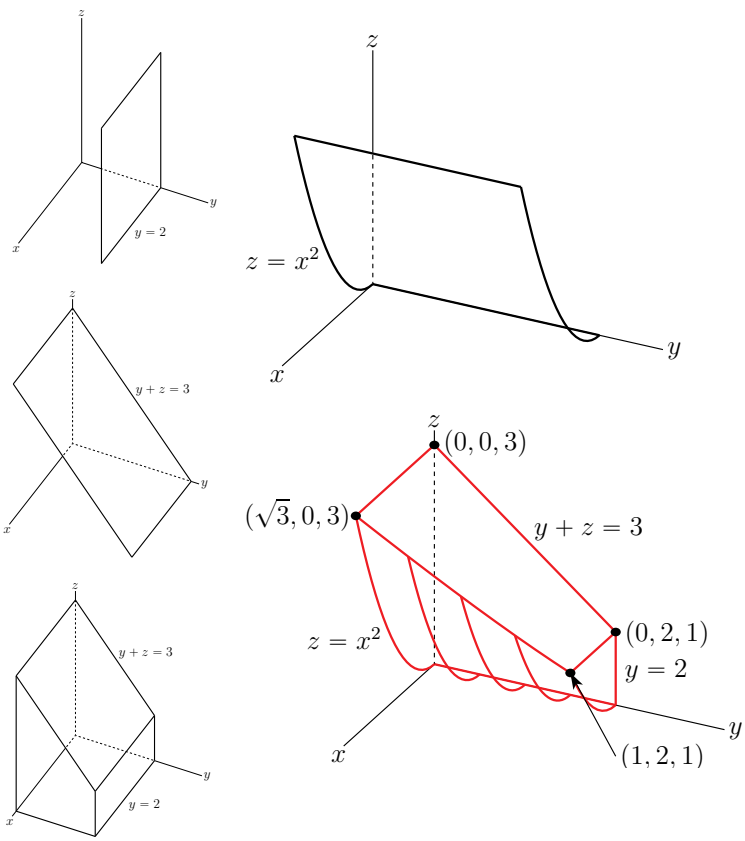


解.



$$
\begin{aligned}
&\int_E f(x,y,z)\,\mathrm{d}V \\
&= \int_{-\sqrt{6}}^{\sqrt{6}} \int_{\frac{y^2}{2}}^{3} \int_{y^2}^{2x} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_{-\sqrt{6}}^{\sqrt{6}} \int_{y^2}^{6} \int_{\frac{z}{2}}^{3} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}z\,\mathrm{d}y \\
&= \int_{0}^{6} \int_{\frac{z}{2}}^{3} \int_{-\sqrt{z}}^{\sqrt{z}} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}x\,\mathrm{d}z
\end{aligned}
$$

例. 若 E 為 $z = x^2$, $y + z = 3$, $y = 0$, $y = 2$ 圍成之區域, 寫出 $\int_E f(x,y,z)\,\mathrm{d}V$ 依積分順序 $z \to x \to y$, $x \to y \to z$, 與 $y \to x \to z$ 之積分式.

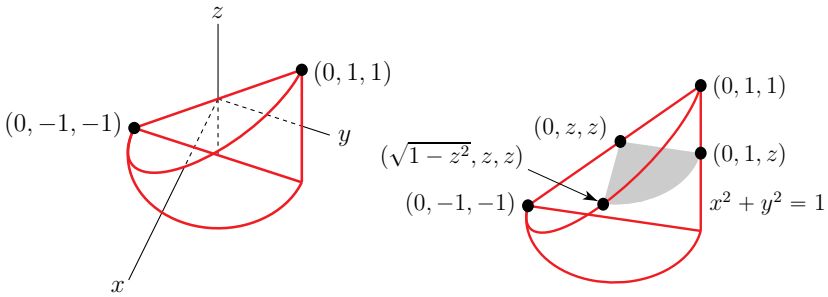


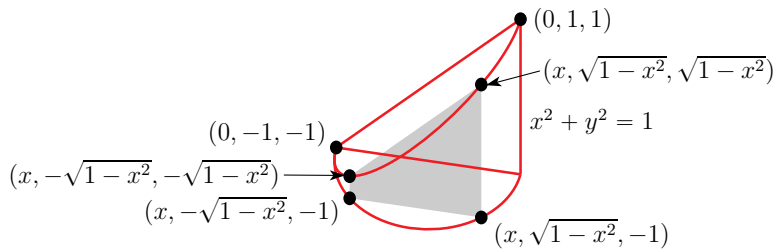
解.

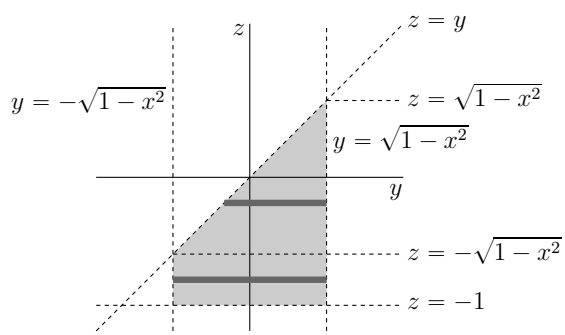
$$
\begin{aligned}
&\int_E f(x,y,z)\,\mathrm{d}V \\
&= \int_0^2 \int_{-\sqrt{3-y}}^{\sqrt{3-y}} \int_{3-y}^{x^2} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_0^1 \int_0^2 \int_{-\sqrt{z}}^{\sqrt{z}} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \\
&\quad + \int_1^3 \int_0^{3-z} \int_{-\sqrt{z}}^{\sqrt{z}} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \\
&= \int_0^1 \int_{-\sqrt{z}}^{\sqrt{z}} \int_0^2 f(x,y,z)\,\mathrm{d}y\,\mathrm{d}x\,\mathrm{d}z \\
&\quad + \int_1^3 \int_{-\sqrt{z}}^{\sqrt{z}} \int_0^{3-z} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}x\,\mathrm{d}z
\end{aligned}
$$

例. 若 E 為 $x^2+y^2 = 1$, $z = y$, $z = -1$ 圍成之區域, 寫出 $\int_E f(x,y,z)\,\mathrm{d}V$ 依積分順序 $z \to x \to y$, $x \to y \to z$, 與 $y \to z \to x$ 之積分式.

解.



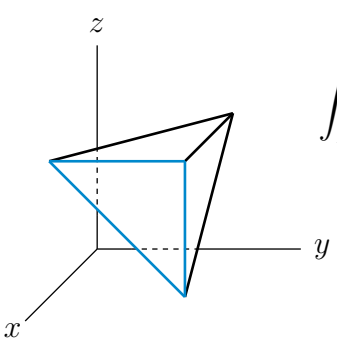




$$
\begin{aligned}
&\int_E f(x,y,z)\,\mathrm{d}V \\
&= \int_{-1}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}}\int_{-1}^{y} f(x,y,z)\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_{-1}^{1}\int_{z}^{1}\int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \\
&= \int_{-1}^{1}\int_{-1}^{-\sqrt{1-x^2}}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}z\,\mathrm{d}x \\
&\quad + \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{z}^{\sqrt{1-x^2}} f(x,y,z)\,\mathrm{d}y\,\mathrm{d}z\,\mathrm{d}x
\end{aligned}
$$

例. 若 E 為由 $x=1$, $y=1$, $z=1$, $x+y+z=2$ 圍成之四面體, 求 $\displaystyle\int_E x\,\mathrm{d}V$.

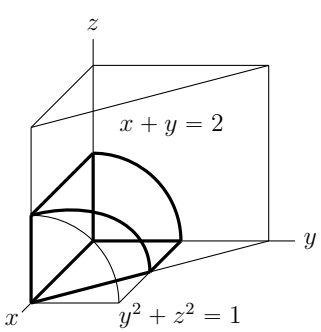
解.



$$
\begin{aligned}
\int_E x\,\mathrm{d}V &= \int_0^1\int_{1-x}^1\int_{2-x-y}^1 x\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x = \int_0^1\int_{1-x}^1 x\,(x+y-1)\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1\int_{1-x}^1 (x^2-x+xy)\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 \left((x^2-x)\,x + x\,\frac{1-(1-x)^2}{2}\right)\mathrm{d}x \\
&= \int_0^1 \left(x^3-x^2+x\,\frac{2x-x^2}{2}\right)\mathrm{d}x = \int_0^1 \frac{x^3}{2}\,\mathrm{d}x = \frac{1}{8}
\end{aligned}
$$

例. 若 E 為第一卦限中由 $y^2+z^2=1$, $x+y=2$, $y=0$, $z=0$ 圍成之區域, 求 $\displaystyle\int_E z\,\mathrm{d}V$.

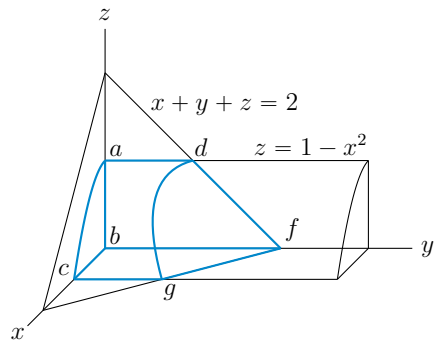
解.



$$
\begin{aligned}
\int_E z\,\mathrm{d}V &= \int_0^1\int_0^{2-y}\int_0^{\sqrt{1-y^2}} z\,\mathrm{d}z\,\mathrm{d}x\,\mathrm{d}y = \int_0^1 \frac{1}{2}\,(1-y^2)(2-y)\,\mathrm{d}y \\
&= \frac{1}{2}\int_0^1 (2-y-2y^2+y^3)\,\mathrm{d}y = \frac{1}{2}\left(2-\frac{1}{2}-\frac{2}{3}+\frac{1}{4}\right) = \frac{1}{2}\cdot\frac{13}{12} = \frac{13}{24}
\end{aligned}
$$

例. 若 E 為由 $x=0$, $y=0$, $z=0$, $x+y+z=2$, $x^2+z=1$ 圍成之區域, 求 $\displaystyle\int_E x\,\mathrm{d}V$.

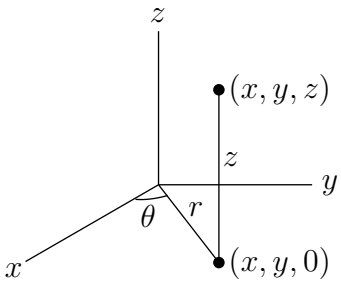
解.



$$\begin{aligned}\int_E x\,\mathrm{d}V &= \int_0^1\int_0^{1-x^2}\int_0^{2-x-z} x\,\mathrm{d}y\,\mathrm{d}z\,\mathrm{d}x = \int_0^1\int_0^{1-x^2}(2-x-z)\,x\,\mathrm{d}z\,\mathrm{d}x\\ &= \int_0^1\left((2-x)\,x\,(1-x^2) - x\,\frac{(1-x^2)^2}{2}\right)\mathrm{d}x\\ &= \frac{1}{2}\int_0^1(-x^5+2x^4-2x^3-2x^2+3x)\,\mathrm{d}x\\ &= \frac{1}{2}\left(-\frac{1}{6}+\frac{2}{5}-\frac{1}{2}-\frac{2}{3}+\frac{3}{2}\right) = \frac{17}{60}\end{aligned}$$

變數變換

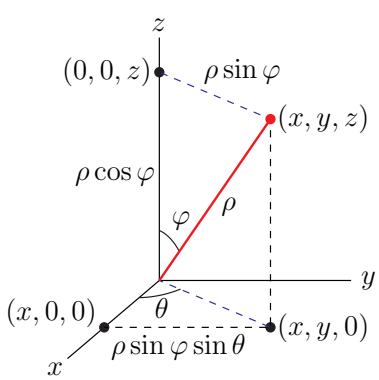
柱面座標



$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \\ z = z \end{cases} \iff \begin{cases} r = \sqrt{x^2+y^2} \\ \theta = \tan^{-1}\frac{y}{x} \\ z = z \end{cases}$$

$$\text{Jacobian } J_{\mathbf{x}}(\mathbf{u}) = \frac{\partial\mathbf{x}}{\partial\mathbf{u}}(\mathbf{u}) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial\theta} & \frac{\partial x}{\partial z} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial\theta} & \frac{\partial y}{\partial z} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial\theta} & \frac{\partial z}{\partial z} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

球面座標

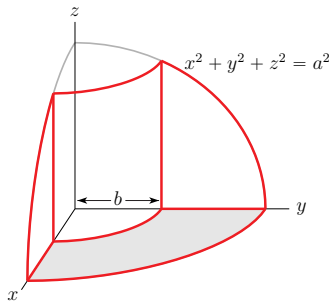


$$\begin{cases} x = \rho\sin\varphi\cos\theta \\ y = \rho\sin\varphi\sin\theta \\ z = \rho\cos\varphi \end{cases} \iff \begin{cases} \rho = \sqrt{x^2+y^2+z^2} \\ \theta = \tan^{-1}\frac{y}{x} \\ \varphi = \tan^{-1}\frac{\sqrt{x^2+y^2}}{z} \end{cases}$$

$$\begin{aligned}\text{Jacobian } J_{\mathbf{x}}(\mathbf{u}) = \frac{\partial\mathbf{x}}{\partial\mathbf{u}}(\mathbf{u}) &= \begin{vmatrix} \frac{\partial x}{\partial\rho} & \frac{\partial x}{\partial\theta} & \frac{\partial x}{\partial\varphi} \\ \frac{\partial y}{\partial\rho} & \frac{\partial y}{\partial\theta} & \frac{\partial y}{\partial\varphi} \\ \frac{\partial z}{\partial\rho} & \frac{\partial z}{\partial\theta} & \frac{\partial z}{\partial\varphi} \end{vmatrix}\\ &= \begin{vmatrix} \sin\varphi\cos\theta & -\rho\sin\varphi\sin\theta & \rho\cos\varphi\cos\theta \\ \sin\varphi\sin\theta & \rho\sin\varphi\cos\theta & \rho\cos\varphi\sin\theta \\ \cos\varphi & 0 & -\rho\sin\varphi \end{vmatrix} = -\rho^2\sin\varphi\end{aligned}$$

例. 半徑為 a 之球中心對稱鑽半徑為 b 之圓孔, $a > b > 0$, 求球剩下的體積。

解.



$$\begin{aligned}(\text{體積}) &= 8\cdot\int_0^{\frac{\pi}{2}}\int_b^a\int_0^{\sqrt{a^2-r^2}} r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta\\ &= 4\pi\cdot\int_b^a r\sqrt{a^2-r^2}\,\mathrm{d}r\\ &= 4\pi\cdot\left(\frac{-1}{3}(a^2-r^2)^{\frac{3}{2}}\Big|_b^a\right) = \frac{4\pi}{3}(a^2-b^2)^{\frac{3}{2}}\end{aligned}$$